

What is Hypothyroidism?

Insufficient thyroid hormone production.

Affects metabolism, growth, and many bodily functions.

Thyroid Physiology: The Hormonal Chain of Command

Hypothalamus releases TRH → stimulates pituitary to release TSH
→ thyroid produces T4 and T3.

T4 (thyroxine) is mostly converted to active T3 in tissues.

Negative feedback loop regulates hormone levels tightly.

Prevalence: 1-4% in general population; up to 10 times more common in women.

Causes of Hypothyroidism

Congenital Hypothyroidism

Primary (95%): Thyroid gland failure (e.g., Hashimoto's thyroiditis, iodine deficiency, surgery).

Secondary: Pituitary dysfunction reducing TSH production.

Tertiary: Hypothalamic dysfunction reducing TRH release.

Classic Symptoms & Signs

Fatigue, weight gain, cold intolerance, constipation, dry skin, depression.

Physical signs: bradycardia, goiter, delayed reflexes, puffiness.

Symptoms often nonspecific; severity varies from mild to life-threatening (myxedema coma).

Diagnosing Hypothyroidism

Blood tests: Elevated TSH with low free T4 indicates primary hypothyroidism.

Subclinical hypothyroidism: Elevated TSH but normal T4.

Thyroid antibodies (TPO) help identify autoimmune causes.

Treatment: Thyroid Hormone Replacement

Levothyroxine (synthetic T4) is the standard treatment.

Dose individualized, adjusted by TSH levels.

Prognosis and Monitoring

Most patients respond well to treatment with symptom resolution.

Regular TSH monitoring essential to avoid under- or overtreatment.

Untreated hypothyroidism can cause serious complications: infertility, heart disease, cognitive impairment.

Congenital Hypothyroidism

Newborn screening critical to prevent intellectual disability.

Early diagnosis and treatment within weeks of birth preserve normal development.

Lifelong monitoring often required.

Conclusion: Managing Hypothyroidism for a Healthy Life

Early recognition and treatment are key.

Patient education on medication adherence and symptom awareness improves outcomes.

Collaboration between patients and healthcare providers ensures optimal thyroid health.